

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Mathematics for Physics II **L-T-P:** 3-0-0 **Credits:** 3

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Partial differential equations (PDE) in Physics: preliminaries; classification of second-order quasilinear equations; elliptic, hyperbolic and parabolic type; First-order equations, characteristics; boundary conditions and types of equations; (8)
Wave equation and d'Alembert's solution; Fundamental solution of the heat equation; the two dimensional Laplace equation; Poisson's equation; the diffusion equation and the wave equation in two dimensions; Dirichlet and Neumann problems; Green's function for PDE; the initial value problem for the wave equation; the method of separation of variables, 3D Laplace equation in spherical polar coordinates; (14)
Group Theory: definitions and nomenclature; examples; finite and infinite groups, continuous and discrete groups, rearrangement theorem; cyclic groups; subgroups and cosets; Cayley's and Lagrange's theorems; conjugate elements and class structure; factor groups; isomorphism and homomorphism; direct product groups; symmetric groups. Non-Abelian groups. Point groups and crystallographic groups, Simple and semi-simple, groups and Sylow theorems. (10)
Representation of finite groups; definition ; unitary representation; Schur's Lemma; orthogonality theorem; reducible and irreducible representations; characters; regular representation; product representation; character table; examples of S_3 and C_{4v} . Elementary ideas of orthogonal and unitary groups and their applications. (10)

Reference Books:

1. Partial differential equations : I.N. Sneddon
2. Partial Differential Equations in Physics: A. Sommerfeld
3. Methods of Mathematical Physics: R. Courant and D. Hilbert
4. Group Theory and its Application to Physical Problems M. Hamermesh

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Partial differential equations (PDE) in Physics: preliminaries; classification of second-order quasilinear equations; elliptic, hyperbolic and parabolic type; First-order equations, characteristics; boundary conditions and types of equations	8
2.	Wave equation and d'Alembert's solution; Fundamental solution of the heat equation; the two dimensional Laplace equation; Poisson's equation; the diffusion equation and the wave equation in two dimensions; Dirichlet and	14

	Neumann problems; Green's function for PDE; the initial value problem for the wave equation; the method of separation of variables, 3D Laplace equation in spherical polar coordinates	
3.	Group Theory: definitions and nomenclature; examples; finite and infinite groups, continuous and discrete groups, rearrangement theorem; cyclic groups; subgroups and cosets; Cayleys and Lagranges theorems; conjugate elements and class structure; factor groups; isomorphism and homomorphism; direct product groups; symmetric groups. Non-Abelian groups. Point groups and crystallographic groups, Simple and semi-simple, groups and Sylow theorems.	10
4.	Representation of finite groups; definition ; unitary representation; Schurs Lemma; orthogonality theorem; reducible and irreducible representations; characters; regular representation; product representation; character table; examples of S_3 and C_{4v} . Elementary ideas of orthogonal and unitary groups and their applications.	10
Total number of hours		42